PRODUCT USER MANUAL

For Baltic Sea Wave Hindcast Product
BALTICSEA_MULTIYEAR_WAV_003_015

Issue: 2.0

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2019/12	All	New product: CMEMS-BAL-PUM-003-015	E. Lindgren L. Tuomi V. Huess	C. Derval
1.1	2021/12		Static files included	V. Huess	C. Derval
1.2	2023/06		Monthly climatology included	H. Kanarik	V.Huess
2.0	2024/11	All	Update of the whole product	H. Kanarik	V. Huess

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GLOSSARY AND ABBREVIATIONS

CF	Climate Forecast (convention for NetCDF)
CMEMS	Copernicus Marine Environment Monitoring Service
ECMWF	European Centre for Medium Range Weather forecast
BAL	Baltic Sea
WAM	WAve Model
NetCDF	Network Common Data Form
PUM	Product User Manual
QUID	Quality Information Document
MDS	Marine Data Store

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1 Summary

This document is the user manual for the CMEMS Baltic Sea Multi Year wave product **BALTICSEA_MULTIYEAR_WAV_003_015**. It provides 43+ years of time series (1980 onward to around 4 months from present) with hourly data.

The wave products are the integrated parameters computed from the total wave spectrum (significant wave height, period, direction, Stokes drift, ...etc), as well as the following partitions: the wind wave, the primary swell wave and the secondary swell wave.

This product covers the Baltic Sea. It is defined on a standard grid at 1 nautical mile (approx. 1.85 km).

The product is organised in files:

- CMEMS_BAL_WAV_MY_${date}.nc containing hourly instantaneous values for the wave variables
- CMEMS_BAL_WAV_MY_AFLUX _${date}.nc containing hourly instantaneous values of sea-air fluxes,
- CMEMS_BAL_WAV_MY_Climatology.nc containing monthly climatology of significant wave height and mean period fields, and
- BAL-MFC_003_010_mask_bathy.nc containing info on the bathymetry and land-sea mask.

The product is published on the CMEMS dissemination server after automatic and human quality controls. Product is available on-line and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.0 convention.

The model system is described in the Quality Information Document (QUID): https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-BAL-QUID-003-015.pdf

I.2 History of changes

Date	Description of changes and impacted product
MARCH 2020	FIRST INTRODUCTION OF A BALTIC SEA WAVE HINDCAST PRODUCT
NOVEMBER 2023	INCLUSION OF MONTHLY CLIMATOLOGY FIELDS FOR VHM0 AND VTM02

NOVEMBER 2024	SYSTEM VERSION UPDATE to WAM4.7, SOURCE TERM PACKAGE UPDATE TO ST4, BATHYMERTY AND OBSTRUCTION FIELD UPATES BASED ON EMODnet BATHYMETRY, INCLUSION OF TWO NEW VARIABLES: VMXL, VCMX AND NEW DATASET CONTAINING SEA AIR FLUXES. UPDATE ON STATIC FILES AND MONTHLY CLIMATOLOGY FIELD RUNS EXTENTED TO COVER YEARS BACK IN TIME FROM 1980 TO 1993

II PRODUCT DESCRIPTION

II.1 General Information about products

Product name	BALTICSEA_MULTIYEAR_WAV _003_015
Geographical coverage	Baltic Sea
Variables	Spectral significant wave height (Hm0) Spectral moments (-1,0) wave period (Tm-10) Spectral moments (0,2) wave period (Tm02) Wave period at spectral peak / peak period (Tp) Mean wave direction from (Mdir) Wave principal direction at spectral peak Stokes drift U Stokes drift V Spectral significant wind wave height Spectral moments (0,1) wind wave period Mean wind wave direction from Spectral significant primary swell wave height Spectral moments (0,1) primary swell wave period Mean primary swell wave direction from Spectral significant secondary swell wave height Spectral moments (0,1) secondary swell wave period Mean secondary swell wave direction from Height of the highest crest Maximum crest trough wave height (Hc,max) Wave Momentum Flux into Sea Water Surface Downward Eastward Stress due to Ocean Viscous Dissipation Surface Downward Northward Stress due to Ocean Viscous Dissipation Charnock coefficient Surface Roughness Length
Available time series	From January 1980 to around 4 months from present
Temporal resolution	Hourly instantaneous values
Horizontal resolution	Approx. 1 nautical miles: For longitude: 1'40'' or 0.028 degrees; For latitude: 1' or 0.017 degrees

Number of vertical levels	surface only
Format	NetCDF4 CF1.0

Table 1: Product Specification

II.2 Details of the datasets

BALTICSEA_MULTIYEAR_WAV_003_015
cmems_mod_bal_wav_my_PT1H-i containing the below listed values with hourly instantaneous values
CMEMS_BAL_WAV_MY_${date}nc: **VHM0** [m] Spectral significant wave height (Hm0) sea_surface_wave_significant_height **VTM10** [s] Spectral moments (-1,0) wave period (Tm-10) sea_surface_wave_mean_period_from_variance_spectral_density_inverse_frequency_moment **VTM02** [s] Spectral moments (0,2) wave period (Tm02) sea_surface_wave_mean_period_from_variance_spectral_density_second_frequency_moment **VTPK** [s] Wave period at spectral peak / peak period (Tp) sea_surface_wave_period_at_variance_spectral_density_maximum **VMDR** [degree] Mean wave direction from (Mdir) sea_surface_wave_from_direction **VPED** [degree] Wave principal direction at spectral peak sea_surface_wave_from_direction_at_variance_spectral_density_maximum

VSDX [m s-1]
Stokes drift U
sea_surface_wave_stokes_drift_x_velo
city

VSDY [m s-1]
Stokes drift V
sea_surface_wave_stokes_drift_y_velocity

VHM0_WW [m]
Spectral significant wind wave height
sea_surface_wind_wave_significant_height

VTM01_WW [s]
Spectral moments (0,1) wind wave period
sea_surface_wind_wave_mean_period

VMDR_WW [degree]
Mean wind wave direction from
sea_surface_wind_wave_from_direction

VHM0_SW1 [m]
Spectral significant primary swell wave height
sea_surface_primary_swell_wave_significant_height

VTM01_SW1 [s]
Spectral moments (0,1) primary swell wave period
sea_surface_primary_swell_wave_mean_period

VMDR_SW1 [degree]
Mean primary swell wave direction from
sea_surface_primary_swell_wave_from_direction

VHM0_SW2 [m]
Spectral significant secondary swell wave height
sea_surface_secondary_swell_wave_significant_height

VTM01_SW2 [s]
Spectral moments (0,1) secondary swell wave period
sea_surface_secondary_swell_wave_mean_period

VMDR_SW2 [degree]
Mean secondary swell wave direction from
sea_surface_secondary_swell_wave_from_direction

VMXL [m]
Height of the highest crest
sea_surface_wave_maximum_crest_height

VCMX [m]
Maximum crest trough wave height (Hc,max)
sea_surface_wave_maximum_height

cmems_mod_bal_wav_my_static: containing the below listed static values

BAL-MFC_003_015_mask_bathy.nc:
mask [1]
Land-sea mask: 1 = sea ; 0 = land
sea_binary_mask

deptho [m]
Bathymetry sea_floor_depth_below_geoid

cmems_mod_bal_wav_my_2km-climatology_P1M-m: containing monthly climatology of the below listed values for the reference period of 01/01/1980–31/12/2023

CMEMS_BAL_WAV_MY_Climatology.nc:

VHM0 [m]
Spectral significant wave height (Hm0)
sea_surface_wave_significant_height

VTM02 [s]
Spectral moments (0,2) wave period (Tm02)
sea_surface_wave_mean_period_from_variance_spectral_density_second_frequency_moment

cmems_mod_bal_wav_my_aflux_PT1H-i: containing the below listed values with hourly instantaneous values

CMEMS_BAL_WAV_MY_AFLUX_${date}.nc:

mflx [W m-2]

Wave Momentum Flux into Sea Water

wave_momentum_flux_into_sea_water **sdes** [Pa] Surface Downward Eastward Stress due to Ocean Viscous Dissipation surface_downward_eastward_stress_due_to_ocean_viscous_dissipation **sdns** [Pa] Surface Downward Northward Stress due to Ocean Viscous Dissipation surface_downward_northward_stress_due_to_ocean_viscous_dissipation **chnk** [1] Charnock coefficient charnock_coefficient_for_surface_roughness_length_for_momentum_in_air **Z0** [m] Surface Roughness Length surface_roughness_length

Table 2: List of the datasets (column 1), of the variable for each dataset (column 2) and their names in the NetCDF files (column 3)

II.3 Product System Description

The Baltic Sea wave hindcast is produced by the Finnish Meteorological Institute (FMI) with the WAM model system. The setup has a spatial resolution of 1 nautical mile.The time series starts on January 1st, 1980 and it includes 1-hourly instantaneous fields of integrated wave parameters from the total spectrum (significant height, period, direction, Stokes drift,...etc), as well as the partition to the wind waves and to the primary and secondary swell waves. Additionally hourly instantaneous sea-air fluxes are provided in separate dataset as well as monthly climatology of means of significant wave height and mean wave period.

The hindcast uses 24 spectral directions and 35 different frequencies between 0.0418 and 1.067 Hz. Meteorological forcing is from ERA5 reanalysis and boundary spectra from ERA5 wave reanalysis at the open boundary at Skagerrak. Ice covered areas are blocked by reading in information on ice concentrations from SMHI/FMI's gridded ice charts. These ice charts are produced by SMHI's and FMI's ice services and based on satellite measurements, observations from ships and coastal measurements sites as well as expert analysis. The resolution and temporal variability of the digitized ice charts varied over the hindcast period: resolution was 3 nmi for the time period 1980-2015, 2.4

nmi for 2015-2017 and 0.25 nmi for 2017 onward, and daily ice charts were available only from 2007 onward. Before 2007, twice a week ice charts were available from SMHI's archives.

Domain Resolution and grid Geographic coverage	BALTIC SEA 9.0318˚E - 30.2080˚E; 53.0083˚N - 65.9081˚N (764 x 775 grid points) App. 1 nautical miles: For longitude: 1'40'' or 0.028 degrees; For latitude: 1' or 0.017 degrees This product covers the Baltic Sea with about 1 nautical mile (approx. 1.85 km resolution).
Algorithm	WAM cycle 4.7
Atmospheric forcing	ERA5
Assimilation scheme	-
Assimilated observations	-
Initial conditions	From previous hindcast
Boundary conditions	Wave spectra from ERA5 wave hindcast
Ice forcing	Ice concentration from SMHI/FMI ice charts.
Surface current forcing	-
Bathymetry	EMODnet Bathymetry

II.4 Processing information

II.4.1 *Time coverage*

Hindcast covers the period of January 1st 1980 to around 4 months from present.

II.4.2 *Time averaging*

The fields are 1-hourly instantaneous at 00, 01, 02, 03, 04, 05, 06, 07, 08, 09, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23 UTC.

III FILE FORMAT

The products are stored using the NetCDF format. To know more about the NetCDF format, please follow this link:

What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:

how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this articles "How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center"

IV.1 File size

DATASET NAME	FILE NAME	DIMENSION [MB]	
		Compressed	Uncompressed
cmems_mod_bal_wav_my_PT1H-i"	CMEMS_BAL_WAV_MY_*datetime*.nc	180	180
cmems_mod_bal_wav_my_aflux_PT1H-i	CMEMS_BAL_WAV_MY_AFLUX_*datetime*.nc	50	50
cmems_mod_bal_wav_my_2km-climatology_P1M-m	CMEMS_BAL_WAV_MY_Climatology.nc	60	60
cmems_mod_bal_wav_my_static	BAL-MFC_003_015_mask_bathy.nc	15	15

IV.2 Remember: scale_factor & add_offset / missing_value / land mask

Real_Value = (Display_Value X scale_factor) + add_offset

The missing value for this product is: -32767s

Land and sea-ice masks are equal to "_FillValue" (see variable attribute on NetCDF file).

IV.3 Structure of file

Examples of the header of output NetCDF files are inserted in VI

V REFERENCES

Quality Information Document
catalogue.marine.copernicus.eu/documents/QUID/CMEMS-BAL-QUID-003-015.pdf

VI. ANNEX

```
netcdf CMEMS_BAL_WAV_MY_2018122401.nc {
dimensions:
    lat = 775 ;
    lon = 764 ;
    time = UNLIMITED ; // (24 currently)
variables:
    float lat(lat) ;
        lat:standard_name = "latitude" ;
        lat:long_name = "latitude" ;
        lat:units = "degrees_north" ;
        lat:axis = "Y" ;
    float lon(lon) ;
        lon:standard_name = "longitude" ;
        lon:long_name = "longitude" ;
        lon:units = "degrees_east" ;
        lon:axis = "X" ;
    int time(time) ;
        time:standard_name = "time" ;
        time:long_name = "time" ;
        time:units = "seconds since 1970-01-01 00:00:00" ;
        time:calendar = "standard" ;
    float VHM0(time, lat, lon) ;
        VHM0:_FillValue = -9999999.f ;
        VHM0:long_name = "Spectral significant wave height (Hm0)" ;
        VHM0:standard_name = "sea_surface_wave_significant_height" ;
        VHM0:units = "m" ;
        VHM0:valid_min = 0.f ;
        VHM0:valid_max = 20.f ;
    float VTPK(time, lat, lon) ;
        VTPK:_FillValue = -9999999.f ;
        VTPK:long_name = "Wave period at spectral peak / peak period (Tp)" ;
        VTPK:standard_name =
"sea_surface_wave_period_at_variance_spectral_density_maximum" ;
        VTPK:units = "s" ;
        VTPK:valid_min = 0.f ;
        VTPK:valid_max = 30.f ;
    float VTM10(time, lat, lon) ;
        VTM10:_FillValue = -9999999.f ;
        VTM10:long_name = "Spectral moments (-1,0) wave period (Tm-10)" ;
```

```
        VTM10:standard_name =
"sea_surface_wave_mean_period_from_variance_spectral_density_inverse_frequency_m
oment" ;
        VTM10:units = "s" ;
        VTM10:valid_min = 0.f ;
        VTM10:valid_max = 20.f ;
    float VTM02(time, lat, lon) ;
        VTM02:_FillValue = -9999999.f ;
        VTM02:long_name = "Spectral moments (0,2) wave period (Tm02)" ;
        VTM02:standard_name =
"sea_surface_wave_mean_period_from_variance_spectral_density_second_frequency_m
oment" ;
        VTM02:units = "s" ;
        VTM02:valid_min = 0.f ;
        VTM02:valid_max = 20.f ;
    float VMDR(time, lat, lon) ;
        VMDR:_FillValue = -9999999.f ;
        VMDR:long_name = "Mean wave direction from (Mdir)" ;
        VMDR:standard_name = "sea_surface_wave_from_direction" ;
        VMDR:units = "degree" ;
        VMDR:valid_min = 0.f ;
        VMDR:valid_max = 360.f ;
    float VHM0_WW(time, lat, lon) ;
        VHM0_WW:_FillValue = -9999999.f ;
        VHM0_WW:long_name = "Spectral significant wind wave height" ;
        VHM0_WW:standard_name = "sea_surface_wind_wave_significant_height" ;
        VHM0_WW:units = "m" ;
        VHM0_WW:valid_min = 0.f ;
        VHM0_WW:valid_max = 20.f ;
    float VTM01_WW(time, lat, lon) ;
        VTM01_WW:_FillValue = -9999999.f ;
        VTM01_WW:long_name = "Spectral moments (0,1) wind wave period" ;
        VTM01_WW:standard_name = "sea_surface_wind_wave_mean_period" ;
        VTM01_WW:units = "s" ;
        VTM01_WW:valid_min = 0.f ;
        VTM01_WW:valid_max = 20.f ;
    float VMDR_WW(time, lat, lon) ;
        VMDR_WW:_FillValue = -9999999.f ;
        VMDR_WW:long_name = "Mean wind wave direction from" ;
        VMDR_WW:standard_name = "sea_surface_wind_wave_from_direction" ;
        VMDR_WW:units = "degree" ;
        VMDR_WW:valid_min = 0.f ;
        VMDR_WW:valid_max = 360.f ;
```

```
    float VPED(time, lat, lon) ;
        VPED:_FillValue = -9999999.f ;
        VPED:long_name = "Wave principal direction at spectral peak" ;
        VPED:standard_name =
"sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
        VPED:units = "degree" ;
        VPED:valid_min = 0.f ;
        VPED:valid_max = 360.f ;
    float VHM0_SW1(time, lat, lon) ;
        VHM0_SW1:_FillValue = -9999999.f ;
        VHM0_SW1:long_name = "Spectral significant primary swell wave height" ;
        VHM0_SW1:standard_name = "sea_surface_primary_swell_wave_significant_height" ;
        VHM0_SW1:units = "m" ;
        VHM0_SW1:valid_min = 0.f ;
        VHM0_SW1:valid_max = 20.f ;
    float VTM01_SW1(time, lat, lon) ;
        VTM01_SW1:_FillValue = -9999999.f ;
        VTM01_SW1:long_name = "Spectral moments (0,1) primary swell wave period" ;
        VTM01_SW1:standard_name = "sea_surface_primary_swell_wave_mean_period" ;
        VTM01_SW1:units = "s" ;
        VTM01_SW1:valid_min = 0.f ;
        VTM01_SW1:valid_max = 25.f ;
    float VMDR_SW1(time, lat, lon) ;
        VMDR_SW1:_FillValue = -9999999.f ;
        VMDR_SW1:long_name = "Mean primary swell wave direction from" ;
        VMDR_SW1:standard_name = "sea_surface_primary_swell_wave_from_direction" ;
        VMDR_SW1:units = "degree" ;
        VMDR_SW1:valid_min = 0.f ;
        VMDR_SW1:valid_max = 360.f ;
    float VHM0_SW2(time, lat, lon) ;
        VHM0_SW2:_FillValue = -9999999.f ;
        VHM0_SW2:long_name = "Spectral significant secondary swell wave height" ;
        VHM0_SW2:standard_name =
"sea_surface_secondary_swell_wave_significant_height" ;
        VHM0_SW2:units = "m" ;
        VHM0_SW2:valid_min = 0.f ;
        VHM0_SW2:valid_max = 20.f ;
    float VTM01_SW2(time, lat, lon) ;
        VTM01_SW2:_FillValue = -9999999.f ;
        VTM01_SW2:long_name = "Spectral moments (0,1) secondary swell wave period" ;
        VTM01_SW2:standard_name = "sea_surface_secondary_swell_wave_mean_period" ;
        VTM01_SW2:units = "s" ;
        VTM01_SW2:valid_min = 0.f ;
```

```
        VTM01_SW2:valid_max = 25.f ;
    float VMDR_SW2(time, lat, lon) ;
        VMDR_SW2:_FillValue = -9999999.f ;
        VMDR_SW2:long_name = "Mean secondary swell wave direction from" ;
        VMDR_SW2:standard_name = "sea_surface_secondary_swell_wave_from_direction" ;
        VMDR_SW2:units = "degree" ;
        VMDR_SW2:valid_min = 0.f ;
        VMDR_SW2:valid_max = 360.f ;
    float VSDX(time, lat, lon) ;
        VSDX:_FillValue = -9999999.f ;
        VSDX:long_name = "Stokes drift U" ;
        VSDX:standard_name = "sea_surface_wave_stokes_drift_x_velocity" ;
        VSDX:units = "m s-1" ;
        VSDX:valid_min = -999.f ;
        VSDX:valid_max = 999.f ;
    float VSDY(time, lat, lon) ;
        VSDY:_FillValue = -9999999.f ;
        VSDY:long_name = "Stokes drift V" ;
        VSDY:standard_name = "sea_surface_wave_stokes_drift_y_velocity" ;
        VSDY:units = "m s-1" ;
        VSDY:valid_min = -999.f ;
        VSDY:valid_max = 999.f ;
    float VMXL(time, lat, lon) ;
        VMXL:_FillValue = -9999999.f ;
        VMXL:long_name = "Height of the highest crest" ;
        VMXL:standard_name = "sea_surface_wave_maximum_crest_height" ;
        VMXL:units = "m" ;
        VMXL:valid_min = 0.f ;
        VMXL:valid_max = 30.f ;
    float VCMX(time, lat, lon) ;
        VCMX:_FillValue = -9999999.f ;
        VCMX:long_name = "Maximum crest trough wave height (Hc,max)" ;
        VCMX:standard_name = "sea_surface_wave_maximum_height" ;
        VCMX:units = "m" ;
        VCMX:valid_min = 0.f ;
        VCMX:valid_max = 50.f ;
// global attributes:
                :Conventions = "CF-1.0" ;
        :source = "FMI-WAM_CMEMS" ;
        :institution = "Baltic MFC, PU Finnish Meteorological Institute" ;
        :cmems_product_id = "BALTICSEA_MULTIYEAR_WAV_003_015" ;
:title = "2D - Hourly Instantaneous “;
        :westernmost_longitude = "9.0138" ;
```

```
        :easternmost_longitude = "30.2080" ;
        :southernmost_latitude = "53.0083" ;
        :northernmost_latitude = "65.9081" ;
        :grid_resolution = "1 nautical mile (ie. 0.0167 degrees northward; 0.0278 degrees
eastward" ;


netcdf CMEMS_BAL_WAV_MY_AFLUX_2017010201 {
dimensions:
    lat = 775 ;
    lon = 764 ;
    time = UNLIMITED ; // (24 currently)
variables:
    float lat(lat) ;
        lat:standard_name = "latitude" ;
        lat:long_name = "latitude" ;
        lat:units = "degrees_north" ;
        lat:axis = "Y" ;
    float lon(lon) ;
        lon:standard_name = "longitude" ;
        lon:long_name = "longitude" ;
        lon:units = "degrees_east" ;
        lon:axis = "X" ;
    int time(time) ;
        time:standard_name = "time" ;
        time:long_name = "time" ;
        time:units = "seconds since 1970-01-01 00:00:00" ;
        time:calendar = "standard" ;
        time:cell_methods = "time: mean" ;
    float chnk(time, lat, lon) ;
        chnk:_FillValue = -9999999.f ;
        chnk:long_name = "Charnock coefficient" ;
        chnk:standard_name =
"charnock_coefficient_for_surface_roughness_length_for_momentum_in_air" ;
        chnk:units = "" ;
        chnk:valid_min = 0.f ;
        chnk:valid_max = 0.32f ;
        chnk:cell_methods = "time: mean" ;
    float z0(time, lat, lon) ;
        z0:_FillValue = -9999999.f ;
        z0:long_name = "Surface Roughness Length" ;
        z0:standard_name = "surface_roughness_length" ;
        z0:units = "m" ;
        z0:valid_min = 0.f ;
```

```
        z0:valid_max = 0.15f ;
        z0:cell_methods = "time: mean" ;
    float mflx(time, lat, lon) ;
        mflx:_FillValue = -9999999.f ;
        mflx:long_name = "Wave Momentum Flux into Sea Water" ;
        mflx:standard_name = "wave_momentum_flux_into_sea_water" ;
        mflx:units = "W m-2" ;
        mflx:valid_min = -10.f ;
        mflx:valid_max = 10.f ;
        mflx:cell_methods = "time: mean" ;
    float sdes(time, lat, lon) ;
        sdes:_FillValue = -9999999.f ;
        sdes:long_name = "Surface Downward Eastward Stress due to Ocean Viscous
Dissipation" ;
        sdes:standard_name =
"surface_downward_eastward_stress_due_to_ocean_viscous_dissipation" ;
        sdes:units = "Pa" ;
        sdes:valid_min = -2.f ;
        sdes:valid_max = 2.f ;
        sdes:cell_methods = "time: mean" ;
    float sdnes(time, lat, lon) ;
        sdnes:_FillValue = -9999999.f ;
        sdnes:long_name = "Surface Downward Northward Stress due to Ocean Viscous
Dissipation" ;
        sdnes:standard_name =
"surface_downward_northward_stress_due_to_ocean_viscous_dissipation" ;
        sdnes:units = "Pa" ;
        sdnes:valid_min = -2.f ;
        sdnes:valid_max = 2.f ;
        sdnes:cell_methods = "time: mean" ;

// global attributes:
        :Conventions = "CF-1.0" ;
        :source = "FMI-WAM_CMEMS" ;
        :institution = "Baltic MFC, PU Finnish Meteorological Institute" ;
        :title = "2D - Hourly Instantaneous air-sea fluxes" ;
        :cmems_product_id = "BALTICSEA_MULTIYEAR_WAV_003_015" ;
        :westernmost_longitude = "9.0138" ;
        :easternmost_longitude = "30.2080" ;
        :southernmost_latitude = "53.0083" ;
        :northernmost_latitude = "65.9081" ;
        :grid_resolution = "1 nautical mile (ie. 0.0167 degrees northward; 0.0278 degrees
eastward" ;
```

```
netcdf CMEMS_BAL_WAV_MY_Climatology.nc{
dimensions:
    lat = 775 ;
    lon = 764 ;
    time =  UNLIMITED ; // (12 currently) ;
variables:
    float lat(lat) ;
        lat:standard_name = "latitude" ;
        lat:long_name = "latitude" ;
        lat:units = "degrees_north" ;
        lat:axis = "Y" ;
    float lon(lon) ;
        lon:standard_name = "longitude" ;
        lon:long_name = "longitude" ;
        lon:units = "degrees_east" ;
        lon:axis = "X" ;
    int time(time) ;
        time:standard_name = "time" ;
        time:long_name = "time" ;
        time:units = "seconds since 1970-01-01 00:00:00" ;
        time:calendar = "standard" ;
    float VHM0(time, lat, lon) ;
        VHM0:_FillValue = -9999999.f ;
        VHM0:long_name = "Spectral significant wave height (Hm0)" ;
        VHM0:standard_name = "sea_surface_wave_significant_height" ;
        VHM0:units = "m" ;
        VHM0:valid_min = 0.f ;
        VHM0:valid_max = 20.f ;
    float VTM02(time, lat, lon) ;
        VTM02:_FillValue = -9999999.f ;
        VTM02:long_name = "Spectral moments (0,2) wave period (Tm02)" ;
        VTM02:standard_name =
"sea_surface_wave_mean_period_from_variance_spectral_density_second_frequency_m
oment" ;
        VTM02:units = "s" ;
        VTM02:valid_min = 0.f ;
        VTM02:valid_max = 20.f ;


// global attributes:
        :_NCProperties = "version=1|netcdflibversion=4.4.1.1|hdf5libversion=1.10.0" ;
        :Conventions = "CF-1.0" ;
```

:source = "FMI-WAM_CMEMS" ;
:institution = "Baltic MFC, PU Finnish Meteorological Institute" ;
:title = "Monthly climatology reference period 1993-2022" ;
:westernmost_longitude = "9.0138" ;
:easternmost_longitude = "30.2080" ;
:southernmost_latitude = "53.0083" ;
:northernmost_latitude = "65.9081" ;
:grid_resolution = "1 nautical mile (ie. 0.0167 degrees northward; 0.0278 degrees eastward" ;